

STAT 479 — Group Project Options (Spring 2026)

This document presents nine group project options for STAT 479. Each project is designed to be completed within a few-week period and aligns with course material through Week 11. Projects emphasize design-based reasoning, causal assumptions, diagnostics, and interpretation rather than heavy coding. Teams should choose one project and work collaboratively to produce a clear, well-structured analysis.

1. Matching as Design: Constructing a Quasi-Experiment from Observational Data

Scientific Question

How can we use matching to approximate the design of a randomized experiment when treatment is not randomly assigned?

Causal Estimand

Average treatment effect (ATE) for a binary treatment in a real dataset.

Project Description

You will be given a real dataset involving a binary treatment (e.g., smoking status, tutoring participation, job training enrollment). The goal is to construct a matched design that mimics a randomized experiment. You must justify confounder selection using subject-matter reasoning, estimate the propensity score, and implement a matching strategy such as nearest-neighbor or full matching. You will evaluate covariate balance using standardized mean differences and graphical diagnostics, discuss overlap, and articulate the assumptions required for identification.

Deliverables

A matched design, balance tables, overlap plots, an ATE estimate, and a discussion of limitations.

Feasibility

Datasets are clean and well-documented; matching is straightforward.

2. Overlap, Trimming, and Effective Sample Size in Practice

Scientific Question

How does limited overlap in covariates affect causal identification, and how can trimming improve estimator stability?

Causal Estimand

ATE in a dataset with imperfect overlap.

Project Description

You estimate the propensity score in simulated datasets and examine the distribution of estimated scores to diagnose overlap. You need to apply trimming rules (e.g., 0.05–0.95, 0.10–0.90) and compute effective sample size under inverse probability weighting. You compare ATE estimates across trimming thresholds and analyze the bias–variance tradeoff. The project highlights the importance of the positivity assumption and the consequences of extreme weights.

Deliverables

Overlap plots, effective sample size calculations, trimmed ATE estimates, and a written interpretation.

3. Subject-Specific Matching Study (Health, Education, or Economics)

Scientific Question

How does matching perform when applied to a domain-specific scientific question with real confounders and context?

Causal Estimand

ATE for a binary treatment in a chosen domain.

Project Description

You choose a domain-specific question, such as the effect of exercise on blood pressure, tutoring on test scores, or job training on earnings. You will be required to identify confounders using domain knowledge, estimate the propensity score, and implement a matching strategy. The emphasis should be on interpreting results in the scientific context and understanding how domain-specific confounding structures influence design choices.

Deliverables

A domain-motivated matched design, balance diagnostics, an ATE estimate, and a contextual interpretation.

4. IPW vs Regression vs AIPW Under Misspecification

Scientific Question

When do inverse probability weighting, regression adjustment, and doubly robust estimators succeed or fail?

Causal Estimand

ATE in a simulated job-training setting.

Project Description

You simulate datasets where earnings depend on education, prior earnings, and motivation. They create four scenarios: correct/incorrect propensity model and correct/incorrect outcome model. You will need to compute IPW, regression, and AIPW estimates and compare bias and variance across scenarios. The project illustrates the double-robust property and the importance of model specification.

Deliverables

A 2×2 table of estimator performance, plots, and a written explanation.

5. Sensitivity Analysis for Unmeasured Confounding (Γ -Based)

Scientific Question

How robust are causal conclusions to the presence of unmeasured confounding?

Causal Estimand

Treatment effect estimate under varying levels of hidden bias.

Project Description

You will be given a matched or weighted dataset and conduct a Rosenbaum Γ sensitivity analysis. You will need to interpret how large Γ must be before the causal conclusion changes. The project emphasizes the limits of observational inference and the importance of assessing robustness.

Deliverables

A table or plot showing how conclusions change with Γ , and a written interpretation.

6. Instrumental Variables in Economics: Quarter-of-Birth or Distance-to-College

Scientific Question

What is the causal effect of an additional year of schooling on earnings, and how does an instrumental variable help identify it?

Causal Estimand

Local average treatment effect (LATE) using an instrument.

Project Description

You use the Angrist–Krueger quarter-of-birth dataset or a distance-to-college dataset. You will be required to justify the IV assumptions (relevance, exclusion, monotonicity), estimate the first stage and 2SLS, and compare IV estimates to OLS and matching estimates. The project highlights why IV estimates differ and what population they apply to.

Deliverables

First-stage results, 2SLS estimates, comparison tables, and an interpretation.

7. Test Statistics in Randomization-Based Inference for Matched Observational Studies

Scientific Question

How does the choice of test statistic influence conclusions in Rosenbaum’s randomization-based inference framework?

Causal Estimand

Sharp null hypothesis of no treatment effect for any unit.

Project Description

You select or are provided with a matched dataset. They implement several test statistics used in randomization-based inference, such as difference in means, Wilcoxon signed-rank, Huber-M, and Hodges–Lehmann aligned rank statistics. You generate randomization distributions under the sharp null, compute p-values, and compare conclusions across statistics. You explore why some statistics detect large effects in a few pairs while others detect small consistent effects across many pairs.

Also, you need to simulate heterogeneous treatment effects to examine design sensitivity in the context of unmeasured confounders.

Deliverables

Randomization distributions, p-value comparisons, and a written explanation of differences.

8. Difference-in-Differences for Policy Evaluation

Scientific Question

Did a real policy intervention (e.g., minimum wage increase, smoking ban, school funding reform) affect an outcome of interest?

Causal Estimand

ATT or ATE using a Difference-in-Differences design.

Project Description

You use publicly available state-level or district-level panel data (I can help you choose one). You will identify treated and control units, check pre-treatment trends, estimate a DiD model, and interpret results in a policy context. You need to discuss the plausibility of the parallel-trends assumption and, if appropriate, explore event-study plots.

Deliverables

Pre-trend plots, DiD estimates, and a policy interpretation.

9. Synthetic Control for a Real Policy Intervention

Scientific Question

How can synthetic control methods be used to evaluate the effect of a policy applied to a single treated unit?

Causal Estimand

Policy effect estimated by comparing the treated unit to a synthetic control.

Project Description

You choose a policy intervention affecting a single unit (e.g., a state implementing a new law). They construct a donor pool, build a synthetic control, assess pre-treatment fit, and interpret post-treatment divergence. They may conduct placebo tests to assess robustness. The project emphasizes graphical interpretation and careful donor selection.

Deliverables

Synthetic control plots, placebo tests (as appropriate), and a written interpretation.
